

The Dark Side of Language Models: Exploring the Potential of LLMs in Multimedia Disinformation Generation and Dissemination

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ABSTRACT

Disinformation - the deliberate spread of false or misleading information poses a significant threat to our society by undermining trust, exacerbating polarization, and manipulating public opinion. With the rapid advancement of artificial intelligence and the growing prominence of large language models (LLMs) such as ChatGPT, new avenues for the dissemination of disinformation are emerging. This review paper explores the potential of LLMs to initiate the generation of multi-media disinformation, encompassing text, images, audio, and video. We begin by examining the capabilities of LLMs, highlighting their potential to create compelling, context-aware content that can be weaponized for malicious purposes. Subsequently, we examine the nature of disinformation and the various mechanisms through which it spreads in the digital landscape. Utilizing these advanced models, malicious actors can automate and scale up disinformation effectively. We describe a theoretical pipeline for creating and disseminating disinformation on social media. Existing interventions to combat disinformation are also reviewed. While these efforts have shown success, we argue that they need to be strengthened to effectively counter the escalating threat posed by LLMs. Digital platforms have, unfortunately, enabled malicious actors to extend the reach of disinformation. The advent of LLMs poses an additional concern as they can be harnessed to significantly amplify the velocity, variety, and volume of disinformation. Thus, this review proposes augmenting current interventions with AI tools like LLMs, capable of assessing information more swiftly and comprehensively than human fact-checkers. This paper illuminates the dark side of LLMs and highlights their potential to be exploited as disinformation dissemination tools.

1. Introduction

The spread of disinformation is not a novel phenomenon, as it can be traced back to the early days of printing. The term "fake news", which has gained considerable prominence in recent years, was first coined in 1925 when an article in Harper's Magazine, titled "Fake News and the Public", lamented how newswires were enabling the rapid dissemination of disinformation (Fake News & the Public | Harper's Magazine, 1925). However, even before this, false and distorted news material was published in reputed newspapers (Soll, 2016). The advent of the internet has fundamentally transformed how information is shared, exacerbating the spread of misleading content. In 2013, the World Economic Forum issued a warning about potential "digital wildfires" which referred to the "viral spread" of intentionally or unintentionally misleading information that could have far-reaching consequences (World Economic Forum, 2012). This overabundance of information has been termed an "infodemic" by the World Health Organization (WHO) (Zarocostas, 2020),

causing confusion among individuals as they struggle to differentiate between true and false information.

As we enter a new era of information warfare, various actors—including state-sponsored entities—are exploiting online social networks to execute targeted disinformation campaigns to manipulate public opinion on specific issues. These campaigns can yield dire consequences for the public, as they can alter opinions and behaviors (Bastick, 2021), particularly concerning critical global events such as major elections (Garrett, 2019). Exposure to false information has been linked to negative impacts on society, such as promotions of anti-vax campaigns (Garrett, 2019), fostering hatred towards political candidates (Balmas, 2014), diminishing trust in media (Ognyanova et al., 2020), as well as creating false memories fabricated policy-relevant events (Murphy et al., 2019).

In recent years, the advancement of Artificial Intelligence Generated Content (AIGC) driven by Generative AI (GAI) technology has gained increasing attention beyond the computer science community (Cao

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et al., 2023). Disinformation has reached a new technical peak for malicious actors, as exemplified by the impactful Deep Fake challenge, where the content's authenticity and generation efficiency are notably high (C. Chen et al., 2023)(T. Zhang, 2022). As the latest technological breakthrough in the field of AIGC, Large Language Models (LLMs) (Zhao et al., 2023a) represented by ChatGPT (Zhang et al., 2023) demonstrate tremendous power in natural language understanding and generation tasks (Zhao et al., 2023a), and could contribute to considerable fields including but not limited to scientific writing (Chen, 2023), education (Baidoo-Anu & Owusu Ansah, 2023), healthcare (Duong & Solomon, 2023; Rao et al., 2023), and other applications. However, raising concerns are induced on the ethical utilization of ChatGPT-like tools as well (Zhuo et al., 2023) regarding the misuse as a potential promotion for disinformation creation and dissemination. With the sophisticated capacities of LLMs, maliciously purposeful actors are enabled to generate convincing and context-aware content (Spitale et al., 2023), which can be easily weaponized to fuel disinformation campaigns (Zhao et al., 2023a). Therefore, this paper reviews the current deficiencies and misuse of LLMs, and in turn introduces the potential correlations to disinformation. As part of clarity, we also showcase an end-to-end pipeline that can be used to create and disseminate disinformation on social media. This development underscores the need for a more proactive and comprehensive approach to combat the spread of false information, incorporating cutting-edge AI-driven tools, multi-stakeholder collaboration, and rigorous research initiatives to address the ever-evolving challenges posed by disinformation in the digital age.

The remainder of this article is structured as follows: Section 2 summarizes the development of AIGC and LLMs, reviews the definition of disinformation, its dissemination objectives, and the role of social media as a channel for spreading disinformation. Section 3 discusses the correlations of LLMs and disinformation from both the technical and human aspects and introduces the potential pipelines that malicious actors may employ to create and disseminate disinformation using LLMs. Section 4 presents existing interventions to counter disinformation and identifies the necessity of using LLMs as further enhancement to adapt or leverage these approaches to tackle the issue effectively. Finally, we conclude the paper in Section 5.

2. Related works

In this section, we focus on the evolution of AIGC and the dynamics of disinformation. We examine the state-of-the-art capacity of AIGC techniques for multimodal content generation and the significant technological shift towards the exploration of natural language based interactive LLMs in the AIGC process and summarize the development of LLMs from the perspectives of technology route, technical breakthrough, and ethical concerns. Concurrently, we investigate the pervasive issues of disinformation, its historical and modern contexts, the

distinct factors influencing its spread, and its multimodal forms. We also describe various actors and their motivations. The above investigations stand as the foundation for the potential intersections between LLMs and disinformation.

2.1. Artificial intelligence generated content (AIGC)

Distinguished from being created by human authors, AIGC refers to the automated creation of large-scale content in a manual-cost saving way based on Generative AI (GAI) techniques (Cao et al., 2023). Developed based on GAI techniques from GAN (Goodfellow et al., 2020) to ChatGPT (C. Zhang et al., 2023), AIGC technically refers to utilizing the GAI algorithms for content generation in the context of human instructions that would guide the model to complete the required task, with minimum human engagement in the production process (Fig. 1).

2.1.1. Generated AI (GAI)

Despite the long history of generative AI models (Douglas A, 2009; Knill & Young, 1997), significant performance improvements and cross-field technical overlap were hardly observed until the advent of deep learning. Attempts at unimodal generative methods within single-modality scope were made in the fields of natural language processing (NLP) (Bengio et al., 2003; Dey & Salem, 2017; Mikolov et al., 2010; S. & J., 1997) and computer vision (CV) (Efros & Leung, 1999; Goodfellow et al., 2020; Heckbert, 1986; Kingma & Welling, 2022) separately, in terms of the distinct forms of features presented in textual and image data as sequences or textures. After going through the explorations for distinct fields, the Transformer architecture (Vaswani et al., 2017) brought the unified framework for generative models in various domains as a technical backbone, with which advanced language models represented by BERT (Devlin et al., 2019) and GPT (Brown et al., 2020) and later image models Vision Transformer (Dosovitskiy et al., 2021) and Swin Transformer (Z. Liu et al., 2021) demonstrated advantages over previous architectures in downstream tasks. Furthermore, beyond unimodal generation, the Transformer enables the models for multimodal generation by learning the connections and interactions from cross-domain data (P. P. Liang et al., 2023), typically represented by the joint vision-language model CLIP (Radford et al., 2021) which integrates visual components with the Transformer and is trained on large-scale text and image data. The emergence of transformer architecture and its derivatives open up a new era of Generative AI and point out the path for large-scale model training. The historical view of the development of Generative AI is given in Fig. 2.

As of now, the family of Generative AI models has grown quite large and can adapt to considerable sorts of tasks based on actual demands. Widely known utilization of unimodal model applications are task-oriented and open-domain chatbots (Adiwardana et al., 2020; Peng et al., 2020) together with machine translation (Jiao et al., 2023; R. Wang et al., 2021) for textual data, and image restoration (Fang et al.,

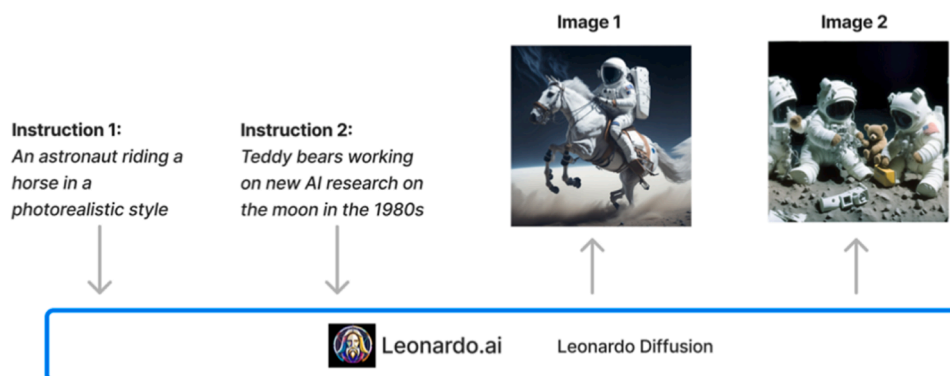


Fig. 1. Examples of AIGC in Image Generation inspired by Cao et al. (2023).

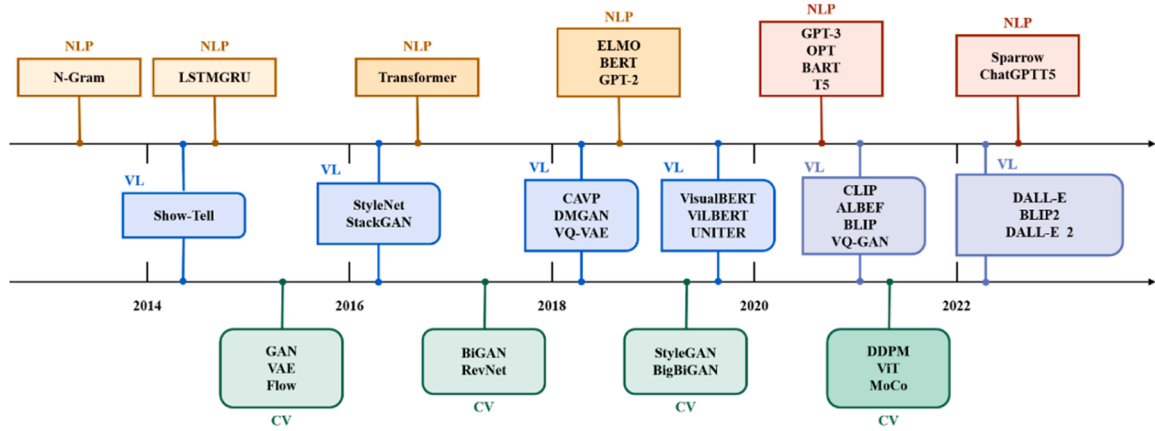


Fig. 2. Historical View of Generative AI Development (Cao et al., 2023).

2022; J. Liang et al., 2021) as well as image editing (Kim et al., 2022; Wallace et al., 2022) for image data. Meanwhile, challenging exploitations of the multimodal model include image captioning (Vinyals et al., 2015) and image generation with given tags (Mansimov et al., 2016) for text-image transform task, speech recognition (Pols, 1976) and speech generation (Ren et al., 2019) for audio-text conversion, and text-guided video generation (Ho et al., 2022; Singer et al., 2022) for the text-video task. Moreover, the state-of-the-art AIGC models have been widely developed into specialized products for practical use (Table 1). Noteworthy, increasing attention is focused on more efficient and better accessible process of content creation on top of high-quality generation in the advancement of AIGC techniques, and the model capacity of understanding human instructions, namely prompt, is in turn becoming the key in achieving a technological breakthrough.

2.1.2. Large language model (LLM)

As a crucial component of AIGC, generative natural language models, notably LLMs exemplified by the popular ChatGPT, have shown exceptional ability in producing realistic and satisfactory content. This enhanced performance is largely attributed to the models' sophisticated techniques in extracting and analyzing intentions from human inputs, as

noted by Cao et al. (2023). The increasing importance of these language models is particularly evident given the textual nature of human instructions, spurring a significant increase in research focused on developing and improving generative language models.

Evolving across the lifespan of individuals and the annals of human history, language stands as the bedrock of human society, facilitating communication and serving as a repository for knowledge documentation (Hauser et al., 2002; Steven, 2003). Tracing back to the seminal work on *intelligent machines* in 1950 (Turing, 1950), a persistent research objective has been the enhancement of machines' capacity to comprehend human language. Based on the evolving comprehension of the nature of language alongside the advancement in computational mathematics, researchers throughout history have continually crafted a succession of ever-optimizing approaches for language modelling.

As the pioneer, Statistical Language Models (SLMs) (Rosenfeld, 2000) originated with the prediction of the subsequent word based on the latest context using Markov assumption, of which the N-gram structures (Cavnar & Trenkle, 1994) are widely applied in both Information Retrieval (Zhai, 2007) and NLP tasks (Bahl et al., 1989). However, SLMs encounter challenges such as dimensional explosion and data sparsity as context information expands.

Table 1
Current Representatives of AIGC Models.

AIGC Task	Modality	Function	Model	Technique	Entity	Year
Text Generation	Unimodal	Text-Text	ChatGPT ¹	GPT 3.5, RLHF	OpenAI	2022
	Multimodal	Image-Text	ChatGPT ²	GPT 4, RLHF	OpenAI	2023
		Speech-Text	USM ³	Self-supervised, Fine-tuning	Google Research	2023
Image Generation	Unimodal	Image-Image	Palette.fm ⁴	Diffusion Model	Google Research	2022
	Multimodal	Text-Image	DALL-E 2 ⁵	CLIP, Diffusion Model	OpenAI	2022
			Midjourney ⁶	–	Midjourney	2022
Speech Generation	Multimodal	Text-Speech	VALL-E ⁷	Neural Codec	Microsoft Research	2023
Video Generation	Unimodal	Unguided	Video-Diffusion ⁸	Diffusion Model	Google	2022
	Multimodal	Text-guided	Make-a-Video ⁹	CLIP, Pseudo-3D Convolutions, Diffusion Model	Meta	2022

¹ www.openai.chat.com.

² www.openai.chat.com.

³ <https://ai.googleblog.com/2023/03/universal-speech-model-usm-state-of-art.html>.

⁴ <https://palette.fm/>.

⁵ <https://labs.openai.com>.

⁶ <https://www.midjourney.com>.

⁷ <https://vall-e.io/>.

⁸ <https://video-diffusion.github.io>.

⁹ <https://makeavideo.studio>.

With the advancement in computational capabilities and the expansion of available data, Neural Language Models (NLMs), initially introduced by Bengio et al. (2003), have significantly matured. A key development in this field was the introduction of distributed word representations, particularly through recurrent neural networks (RNNs), as demonstrated by Mikolov et al. (2010). These models, utilizing aggregated context features, were instrumental in predicting subsequent words in a sequence. This approach was further refined with the development of the Word2Vec model by Mikolov et al. (2013), which marked a substantial contribution to the representation learning method in natural language processing.

Continuing along the trajectory of representation learning, Pre-trained Language Models (PLMs) have emerged as effective tools for various NLP tasks. Examples include ELMo (Peters et al., 2018), which is based on pre-training a bidirectional LSTM (Hochreiter & Schmidhuber, 1997) network to capture context-aware word representations, and BERT (Devlin et al., 2019), which employs pre-training on bidirectional Transformers (Vaswani et al., 2017) with large-scale unlabeled corpora, followed by fine-tuning on downstream tasks. These models have pioneered the 'pre-training to fine-tuning' learning paradigm, inspiring extensive follow-up work on model architecture (Lewis et al., 2020; Radford et al., 2018) and pre-training strategies (Liu et al., 2019; Sanh et al., 2022; Wang et al., 2022)."

Continuing along the trajectory of representation learning, Pre-trained Language Models (PLMs) have emerged as effective tools for various NLP tasks. Examples include ELMo (Peters et al., 2018) based on pre-training a bidirectional LSTM (Hochreiter & Schmidhuber, 1997) network to capture context-aware word representations, and BERT (Devlin et al., 2019) based on the pre-training on the bidirectional Transformer (Vaswani et al., 2017) with large-scale unlabeled corpora, followed by fine-tuning on downstream tasks. Such models pioneer the "pre-training to fine-tuning" learning paradigm, inspiring extensive follow-up work on model architecture (Lewis et al., 2020; Radford et al., 2018) and pre-training strategies (Liu et al., 2019; Sanh et al., 2022; Wang et al., 2022).

In the research space of PLMs, researchers noted the impact of model scale on language model performance and the Empirical Scaling Laws for neural language models proposed in the summative work by ((Kaplan et al., 2020) clarifies the optimal return of much larger models considering factors model size, dataset size and computation cost. These insights hint at the imminent era of Large Language Models (LLMs).

The evolutionary trajectory of LLMs becomes evident through the lineage of the GPT family developed by OpenAI since 2017 (Table 2). The initial offerings, GPT-1 (Radford et al., 2018) and GPT-2 (Radford et al., 2019) demonstrated the ability to learn long-range dependencies from diverse corpora of contiguous text, handling long stretches and zero-shot tasks. Marking a technological leap forward, GPT-3, the third iteration, is a breakthrough for the GPT family, being 100 times larger than its predecessor GPT-2 and trained on a massive dataset comprising

500 billion words from the Common Crawl. GPT-3 introduces the novel concept of In-Context Learning (ICL) (Dong et al., 2023) enabling the model to learn few-shot and zero-shot tasks without model tuning. It also demonstrates generalization capacity in domain adaption and reasoning. Building on the success of GPT-3, subsequent enhancement has been made on the logical thinking and reasoning capacity of the GPT-3.5 model and the multimodal capacity for the GPT-4 model (OpenAI, 2023). As the milestones in the realm of language models, the GPT series not only outperforms the state-of-art benchmarks across a range of classic NLP tasks but also influence the trajectory of human-centered AI systems, steering such towards the realization of Artificial General Intelligence (AGI) (Bubeck et al., 2023).

A notable departure from the predecessors, GPT-3 and subsequent models bring about a substantial shift in both model size and data scale during training. This change has led to the categorization of such models as LLMs by the research community (Hoffmann et al., 2022; Shanahan, 2023). Speaking beyond training, the emergence of LLMs, represented by remarkable applications like ChatGPT, has also begun to reshape how humans engage with language models.

2.1.3. ChatGPT and its technical architecture

As the representative of LLMs, the release of ChatGPT (Zhang et al., 2023) by OpenAI in 2022 attracted numerous users flocking to the unprecedented language model, the number of which exceeded 100 million within only two months, surpassing the proliferation rate of other social products (Neethling, 2023). Astonished by the superior natural language understanding capacity and strong knowledge base, it has been receiving considerable media coverage (Derico, 2023; Hughes, 2023) and the attention of stakeholders from industry (Mehdi, 2023) and academia (Rudolph et al., 2023; Sun, 2022). Being a fine-tuned version of GPT-3.5 and GPT-4, ChatGPT is presented in the form of a dialogue system while following the properties of LLMs obtained in the training process as being built on the unified framework.

Early GPTs. The Transformer (Vaswani et al., 2017) is the acknowledged backbone for the advanced generative AI models, using the self-attention mechanism as the core. Given an input text, self-attention indicates the unique weights of the component words based on contextual dependencies and relationships of the text. The advanced multi-head attention allows the model to capture information from different representation subspaces and improves the understanding capacity.

Based on the Transformer architecture, generative pre-training (Radford et al., 2018) is then proposed to train unsupervised language model on large-scale unlabeled data to learn language patterns and representations. Autoregressive methods (Bengio et al., 2003; Uriu, 2016) in the context of language modeling predicts the following words sequentially according to the previous words as input, based on which the model could learn a joint distribution of the text as a joint product of the previous conditional distributions in the word variable sequence:

Table 2

The GPTs (Generative Pre-training Transformers) family.

Model	Parameters	Data	Core Technique	Capacity	Year
GPT-1	0.117 Billion	BookCorpus ¹⁰	Pre-training Transformer (Decoder-only)	Zero-shot Learner	2018
GPT-2	1.5 Billion	WebText ¹¹	Unsupervised learning Probabilistic Formulation	Multitask Learner	2019
GPT-3	175 Billion	Common Crawl ¹²	GPT-2	In-context Learning	2020
GPT-3.5	–	Code Crawl (M. Chen et al., 2021)	GPT-3	Chain-of-thought	2022
		Human Preference	Supervised Fine-tuning RLHF	Human Alignment	
GPT-4	–	–	–	LLM Risk Mitigation Multimodality	2023

¹⁰ <https://huggingface.co/datasets/bookcorpus>.

¹¹ <https://huggingface.co/datasets/openwebtext>.

¹² <https://commoncrawl.org/>.

$$p_{\theta}(x) = p_{\theta}(x_1)p_{\theta}(x_2|x_1)\dots p_{\theta}(x_n|x_1, x_2, \dots, x_{n-1}) \quad (1)$$

The autoregressive model can be interpreted as a feedforward network incorporating all the preceding contextual information, which forms the foundation of GPT models. In an early stage, models from GPT-1 to GPT-3 are built based on the decoder-only Transformer architecture below and further fine-tuned on specific tasks (Fig. 3).

GPT-3.5 and GPT4. On top of the GPT-3 model, the GPT-3.5 is extended with supervised finetuning, and reinforcement learning with human feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022). Capturing a large number of language patterns and representations, GPT-3.5 is then expected to logically generate suitable content in the specific contexts. Therefore, pairs of prompts and the corresponding answers written by human labeler are constructed as the dataset for the supervised fine-tuning (Fig. 4), namely instruction tuning (Wei et al., 2022), and large-scale code data are used for model enhancement as well (Chen et al., 2021). Furthermore, considering a higher-level human-machine alignment and the creativity of the model to avoid boilerplate answers, the RLHF is applied to the model, in which a reward model is trained firstly based on the data given by human labeled that indicate the rank of the generated contents for a single prompt, and is then used to give the rewards for the outputs to update the model policy using Proximal Policy Optimization (PPO) (Schulman et al., 2017) in the large loop. Due to the data label-independent property, RLHF punishes and rewards the model behavior and plays the significant role in the breakthrough of GPT-3.5 (Fig. 4). Beyond unimodality, the latest GPT-4 model can handle multimodal tasks from image and textual input to textual output process multimodal input. Despite of the unreleased technical details, GPT-4 demonstrates higher capacity of understanding and human alignment, indicating the continuation of the previous but enhanced training framework.

ChatGPT. As the fine-tuned versions of GPT-3.5 and GPT-4, ChatGPT constructs the user interface and highlights the characteristic of LLM as dialogue system. With the better accessibility of human-machine interaction, the major approach for task-oriented model adaption are found to be In-Context Learning (ICL) (Dong et al., 2023) and Chain-of-Thought Prompting (CoT) (Wei et al., 2023). Improving the model performance with distinct prompting formats, ICL prompts consist of the objective task description and a series of task-solution samples as the learning material for the model to learn in such context, while CoT prompts integrate intermediate reasoning steps to the final outputs (Fig. 5).

Despite of the similarity of supervised instruction tuning and prompt engineering to be based on human instructions, it is worth mentioning that the former fine-tunes the architecture of model while the latter prompts it for context-specific utilization only.

So far, in the course of the engagement in the study of the LLMs

represented by ChatGPT, researchers have gained insights into the significantly distinct features of LLMs in comparison to the predecessors. Firstly, due to the complete black-box nature of LLMs, understanding and steering their operations across various tasks necessitate interaction through instruction tuning (Chan, 2023; Zhao et al., 2023a), which is referred to as prompt engineering (Liu et al., 2023; Shieh, 2023) as well. Secondly, with the foundational model and training framework in place, further improvement rely increasingly on the acquisition of training data for both supervised tuning and human preference alignment. This shift in focus places a significant emphasis on the engineering of large-scale data for AI scientists. Finally, along with the extensive growth in model scale, there comes the emergent ability (Wei et al., 2022b), which refers to the unpredictable generations of LLMs including instances containing unseen data. This ability not only piques the interests of researchers across diverse domains (Zhao et al., 2023a), but also raises concerns about potential adverse consequences as highlighted in early research (Ji et al., 2023; Wei et al., 2022b). Such concerns point to potential risks of misuse associated with LLMs (Zhao et al., 2023b), elevating the significance of truthfulness, transparency, and accountability in LLM-based AI technologies to an unprecedented level.

2.2. Disinformation

In the aftermath of the 2016 US presidential election, the concept of “fake news” garnered significant attention from both the media and academic spheres. This term shares similarities with other forms of misleading information, including misinformation and disinformation, each encompassing narratives, theories, or opinions that propagate rapidly through social media or online media channels. Disinformation has a long-standing presence throughout history, playing a role in the rise of Roman emperors who utilized messages on coins as a form of mass communication (Turner et al., 2010), and featuring prominently in Nazi propaganda that heavily relied on the printed press, radio, and cinema (Herf, 2005). However, with the rise of digital platforms and the internet, the reach of disinformation campaigns has become substantially more extensive and pervasive. The ease of content creation, rapid dissemination, and potential for micro-targeting specific audiences have significantly amplified the impact of these campaigns (Mihindukulasuriya, 2020). Moreover, the interactive nature of digital platforms enables disinformation to spread through a combination of algorithmic recommendations and user-driven sharing, which often reinforces echo chambers (Flaxman et al., 2016) and exacerbates social polarization (Mahapatra & Plagemann, 2019).

Misinformation is defined as “false or misleading information” and disinformation is defined as “false information that is purposely spread to deceive people” (Lazer et al., 2018). The primary distinctions between

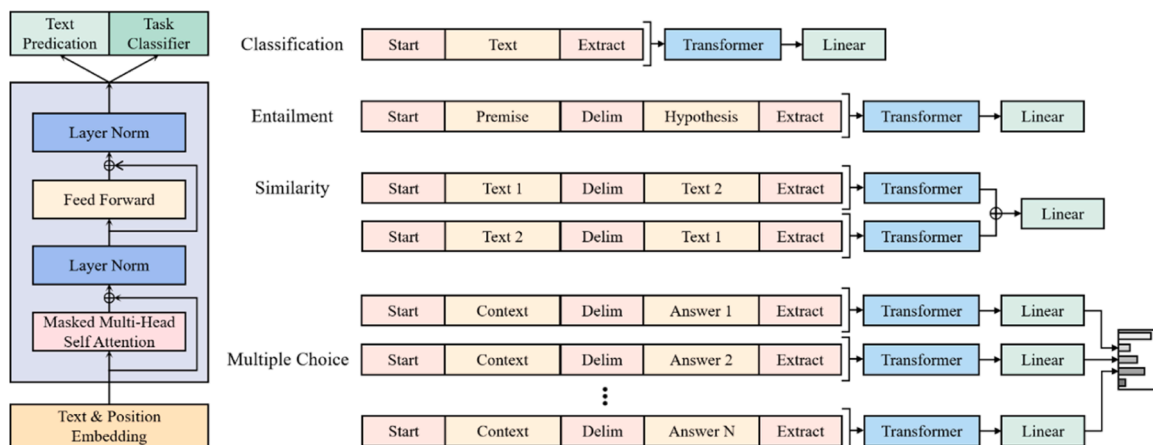


Fig. 3. Training architecture and tuning objectives in GPTs (Radford et al., 2018).

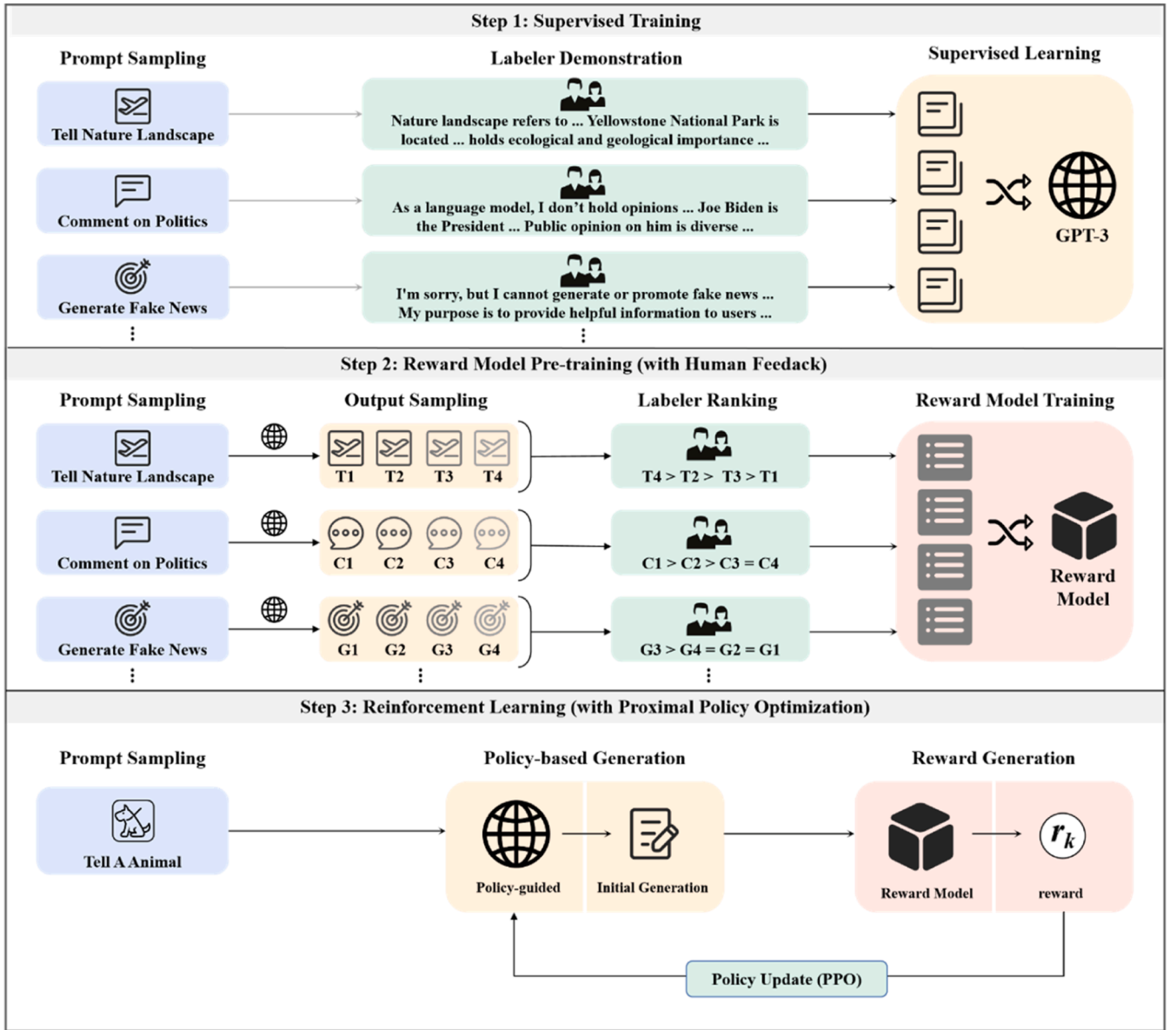


Fig. 4. Supervised fine-tuning and RLHF workflow of GPT-3.5.

these terms lie in the intent behind their dissemination and the way they are spread. Although the term “Fake news” has garnered significant attention, its lack of definitional rigor makes it a problematic label. Lazer et al. (2018) described it as fabricated information that mimics news media content, but this definition fails to capture the phenomenon’s complexity, which can encompass satire and deliberately misleading content aimed at achieving political or other objectives (Caucaso, 2017). A report by the UK parliamentary committee concluded that “fake news” is bandied around with no clear idea of what it means, or agreed definition (House of Commons, 2019). The term has adopted various meanings, even being used to describe statements that a reader dislikes or disagrees with. Considering the politicization of the term by influential figures seeking to discredit certain news media (Vosoughi et al., 2018), we avoid using “fake news” throughout this paper.

Disinformation carries substantial impact primarily due to the cognitive, social and emotional factors of an individual. Individuals are predisposed to certain biases such as confirmation bias (Murungi, Yates, Puro, Yu, & Zhan, 2019), which propels individuals to embrace information that aligns with their existing beliefs regardless of the ground

truth. This is compounded by the continued influence effect, where even after debunking the disinformation, it still shapes people’s opinion and reasoning, adversely affecting the decision making process (Lewandowsky & van der Linden, 2021). Social and emotional states also intensify the reach of disinformation. Individuals may uphold disinformation that resonates with their social group’s beliefs or that induces strong emotional responses, particularly fear or anger further affecting individuals (Calvillo et al., 2020). The current era of digital platforms allows to amplify disinformation’s reach and impact. These platforms rapidly propagate misinformation to vast audiences and create “Echo Chambers” (Flaxman et al., 2016) that primarily expose users to information that aligns with their beliefs, further solidifying their convictions to the disinformation and building resistance to its correction. Efforts to tackle disinformation frequently fall short due to the aforementioned cognitive, social, and emotional barriers, allowing the influence of disinformation to endure over time (Ecker et al., 2022). Consequently, disinformation significantly impacts individuals and their decision-making, societal harmony, and public trust in journalist media, underscoring the importance of effective countermeasures.

Beyond the epistemic analysis of disinformation impact, distinct

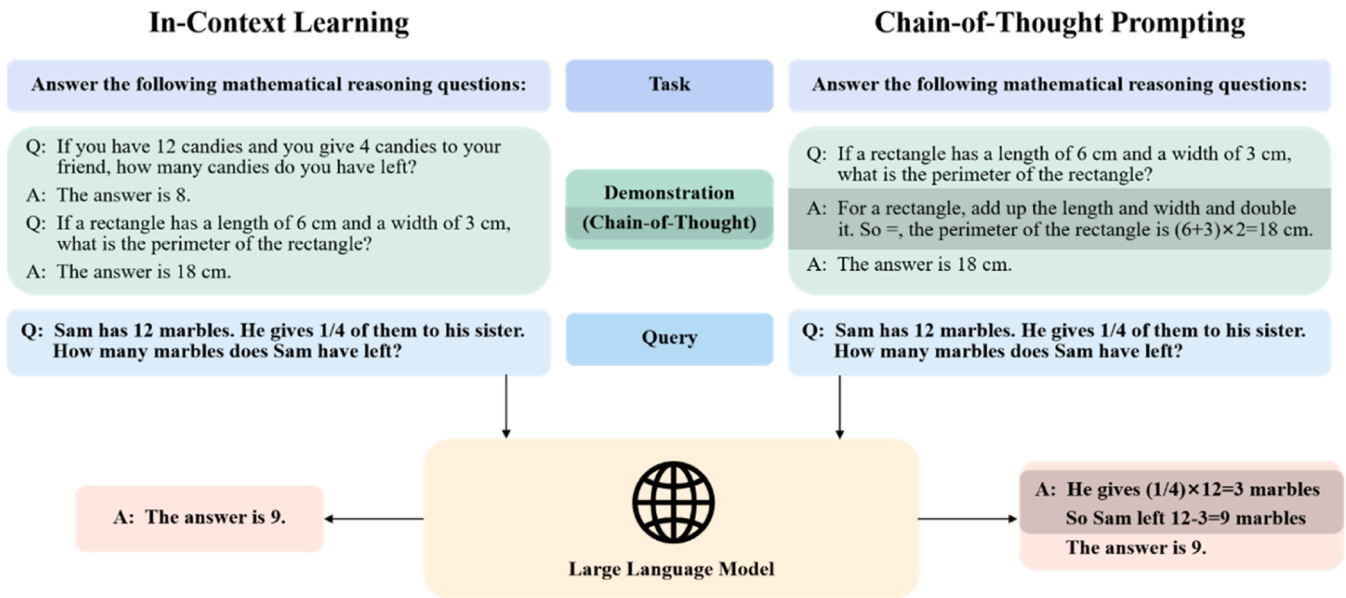


Fig. 5. In-context learning and chain-of-thought prompting (Zhao et al., 2023).

presentations of disinformation are affecting audience to varying degrees on different scales. Early study states the basis of visual disinformation strategy to be the convincing nature of image or video evidences as explicit representations of reality, compared to abstract forms of communications such as texts (Messaris & Abraham, 2001). President Trump claimed the crowd of his inauguration to be the biggest in history with edited photos against reality, which was proved infamous disinformation by other images on Twitter (Swaine, 2018). Multimodality is becoming the common property of current online disinformation (Harcup & O'Neill, 2017). Hameleers et al. (2020) studies the impact degree of multimodal and textual disinformation on people measured by the qualified credibility score through social experiment and makes the conclusion that multimodal disinformation is considered to be more credible than texts that audience are less suspicious of, though visual editing has been an alarming open problem to the public. Further, the advancement of AI techniques is boosting the disinformation issue (Bontridder & Poulet, 2021) by not only increasing the opportunity to create unidentifiable AIGC (Kim et al., 2022; Wallace et al., 2022), but also allowing multimodal generation for fictitious content with human instructions (Mansimov et al., 2016; Singer et al., 2022). As a category of multimodal disinformation, the Deep Fake issue induced by AIGC has been widely investigated on its motivations, impacts on various aspects and detection methods (Kertysova, 2018; Whyte, 2020).

2.3. Dissemination of disinformation

With the rise of digital platforms, the dissemination of disinformation has been propelled into another whole new level. We summarized a handful of actors that help in the propagation of disinformation as follows:

- **Activist or Political Organization:** Numerous organizations disseminate false information with the intention of promoting their own agendas, undermining rival organizations, or advancing a specific narrative to the public. For example, a group of activists claimed that the coronavirus vaccines were a “*medical fraud*” and tried to disseminate false information to manipulate individuals about vaccines (Frenkel, 2021).
- **Bots (Davis et al., 2016):** In the realm of false information, bots are software programs that operate as part of a botnet, controlling the online activity of numerous fake accounts to disseminate misleading

content. Botnets typically consist of a substantial number of fake accounts used to propagate false information across the internet. Often, third-party organizations employ botnets for profit, with various motives driving their deployment of false information. It is important to note that different types of bots with varying capabilities exist; for example, some bots solely repost and promote content (e.g., through vote manipulation on platforms like Reddit), while others generate “*original*” content.

- **Criminal and Terrorist Organizations (Piazza, 2022):** Criminal gangs and terrorist organizations exploit digital platforms to disseminate false information in pursuit of their objectives. A notable example is the ISIS terrorist organization, which spreads misleading information on digital platforms for propaganda purposes (Piazza & Guler, 2021). Their tactics include disseminating ideologically charged messages aimed at recruitment. This poses a significant threat to communities, as there have been multiple instances of individuals from European countries being recruited by ISIS, ultimately leading to the perpetration of terrorist acts.
- **Governments:** Governments have engaged in the dissemination of false information for a variety of purposes. In recent times, with the widespread adoption of the internet, governments turn to social media to manipulate public opinion on specific issues. Moreover, there are reports suggesting that foreign governments disseminate false information concerning other countries to influence public opinion on matters relating to those nations. Notable examples include the alleged involvement of the Russian government in the 2016 US elections (Abram, 2019) and the Brexit referendum (Ellehuus, 2020).

Furthermore, to delve deeper into why a certain disinformation is created, there presents the necessity of understanding the motivations of why actors would choose to create it in the first place. We hereby summarize four major motivations for the spread of disinformation. They are:

- **Financial Gain** - Numerous actors within the false-information ecosystem pursue popularity and financial gain for their organizations or websites. They often achieve this by disseminating false information that drives traffic to their websites, leading to higher advertising revenue generated at the expense of manipulated users and potentially harming them (Debra Kelly, 2020). Examples of such

tactics include the use of clickbait techniques and the creation of fabricated news stories, which aim to increase views of articles from fake news sites disseminated via Digital Platforms (Zannettou et al., 2019).

- **Political Gain** – This motive pertains to the intention of deceiving others in order to influence their decisions or manipulate public opinion on specific issues. Two general categories can be identified within this motive: (1) seeking leverage or followers (*power*), and (2) attempting to alter public norms by spreading false information. False information could also be spread to sow confusion and create an environment of mistrust among individuals (Ognyanova et al., 2020). This is particularly concerning in the realm of politics (Mahapatra & Plagemann, 2019), where individuals disseminate false information to bolster their public image or to tarnish the reputation of opposing politicians, especially during election periods. For example; India's largest democratic party have been found to employ Bots and troll to spread 'Fake News' on Twitter (Regina Mihindukulasuriya, 2020).
- **Ideology** – A significant number of users possess strong convictions regarding a particular idea, organization, or entity. This passion can impact their judgment, leading to the dissemination of false information (Calvillo et al., 2020). Specifically, such fervent users may become blinded by their ideology, causing them to perceive and accept false information as accurate, thus contributing to its overall propagation.
- **Trolling** – Online trolls are usually individuals who are motivated to spread disinformation on digital platforms for their amusement (Trottier, 2020). Their actions at times, can cause significant panic and harm to society and therefore should not be underestimated or dismissed.

Understanding the motivations behind disinformation creation and dissemination provides a crucial background against which we examine the potential role of ChatGPT in the realm of disinformation. The profound capabilities of this language model, combined with the underlying motives, would create a favorable environment for malicious exploitation. This is indicated by the aforementioned technological prowess of ChatGPT as a potential catalyst accelerating the spread of disinformation. Subsequently, we will delve into a comprehensive exploration of the technical provenance of ChatGPT deficiencies and security vulnerabilities and introduce its inherent connection to disinformation.

3. Large language model and disinformation

Generative AI has been a significant concern, particularly in its association with disinformation. Malicious actors have exploited these technologies for creating convincing and hard-to-detect fake content, thereby advancing their deceptive agendas (Whyte, 2020; Zhang, 2022). The capabilities of ChatGPT, a prominent LLM, have been both impressive and concerning. Extensive research focusing on AI safety and trustworthiness has delved into the inherent flaws and ethical challenges posed by LLMs. Studies have highlighted the potential for LLMs to generate incorrect, biased, or harmful content in response to certain inputs (Borji, 2023; Zhuo et al., 2023). This propensity exacerbates the issue of disinformation. The effectiveness of LLMs like ChatGPT in shaping (dis)information is influenced by the training paradigm used and human interactions with these systems.

3.1. Relating large language model and disinformation

Based on the large-scale training architecture and human-alignment tuning, LLMs represented by ChatGPT demonstrate strong capacities in a wide range of domains (Baidoo-Anu & Owusu Ansah, 2023; T.-J. Chen, 2023; Duong & Solomon, 2023) but is recognized to have a series of deficiencies as well (Borji, 2023; Zhao et al., 2023b). With the characteristics from both sides, LLM could play the key role, directly or

indirectly, in the process of disinformation creation and dissemination.

3.1.1. Provenance from training data

With the strong demand of large-scale data for model pre-training, the source and processing method deeply influence the functioning of LLMs from the perspective of information quality.

Data Quality. As the knowledge foundation of LLMs, the pre-training on the low-quality corpus including toxic, biased, noisy, and duplicated data could result in harmful model output (Hernandez et al., 2022; Rae et al., 2022). Despite of the pre-processing of training data, research still reveals the possibility of LLMs to produce pornographic, vulgar, violent, and biased content due to the hardship of complete training data review and the inevitability of the biased human-generated feedback data (Hartmann et al., 2023; McGee, 2023), which can be the origins of disinformation. Specifically, the large-scale data collection process of LLMs could result in the possible violation of data privacy, including personal privacy breach and organizational confidentiality leaks, which can be transformed into malicious disinformation targeting personal reputation or causing social unrest.

Data Distribution. Pre-trained on a large-scale mixture of sources, LLMs acquire a wide range of knowledge from various domains. However, the distribution of training data across domains affects the generalization capacity of the model. Imbalanced training data could result in poor performances of the model in specialized domains, such as medical treatment applications (Borji, 2023) and the numerical computations (Qian et al., 2022). Such deficiency in data can result in unexpected model generation containing factual or reasoning error due to knowledge gaps or conceptual confusion and in turn produce domain-specific disinformation to mislead the public (Eggmann et al., 2023).

Data Timeliness. Relying on the latest knowledge, the performance of ChatGPT on timely tasks is dictated by the timeliness of training data, namely knowledge recency (Zhao et al., 2023b). Enquiries about time-sensitive information can not only result in possibly identifiable fallacies but generating content containing competitive truth not up to date with current as well (Farris, 2018), which is more dangerous to the public as the misleading but truthful disinformation. Existing research tackling the problem still remains superficial on the proper methods to integrate new factual knowledge as data in the training process (Dai et al., 2022; Meng, Bau, Andonian, & Belinkov, 2022).

3.1.2. Provenance from model training

Illustrating that training data provides the learning material for the model, then the training of the model builds the capacity of knowledge learning and utilization, which largely determines the model outputs. The complex model training process indicates the strong capacities while implying the potential risks of LLMs as well.

Generative Pre-training. Language models generate response by predicting the following words based on the previous context (Radford et al., 2018). Such property technically determines ChatGPT to be a powerful statistical model at best, which lacks rational human thinking to pass Turing test and has no consciousness (Hanna, 2023). Despite the remaining controversy on the definition of self-awareness, it is arguably stated at present that ChatGPT is not capable of a global view to perform inference in a meaningful way or interpret psychology and behavior (Borji, 2023). This indicates that ChatGPT is unaware of the harmful contents and fake narratives it learns from massive data and relies totally on human regulation. Also, the unidirectional modelling method of the GPTs technically decides the non-fully contextual nature and is thus not able to neither control the authenticity and logic of the generation, nor make necessary verification, indicating the high risk of harmful model output and the produce of disinformation.

Scaling. Scaling has been clarified to be the key factor of the capacity leap of LLMs (Kaplan et al., 2020). With the extremely increased training scale, what LLMs demonstrates in its generation capacities and deficiencies are the appearance of new aspects or the enhancement in

existing aspects compared to the predecessors:

- **Emergent Ability** refers to “the ability that are not present in small models but arise in large models” (Wei et al., 2022b), which indicates the significant performance improvement as the training scale exceeds a specific threshold in a variety of tasks. As the representatives, **in-context learning** and **instruction following** abilities enable ChatGPT to generate high-quality content when provided with a single natural language instruction or together with several task demonstrations without additional model tuning, namely zero-shot and few-shot learning (Brown et al., 2020; Wei et al., 2022a). With such considerable generalization capacity, ChatGPT acquires the strong capacity in understanding and reacting to human instructions and can associate unseen data according to its knowledge base and show great creativity in response.
- **Reasoning** in the context of LLM refers to the coherent and logical response of the model (Wei et al., 2023). **Step-by-step reasoning** is the other representative **emergent ability** of ChatGPT. For the complicated tasks requiring step-by-step solutions, given the reasoning process in the form of **chain-of-thought** prompt, ChatGPT could figure out the answer with the simulated mechanism through a series of logically connected steps (Wei et al., 2023). The above emergent abilities set up the foundations of ChatGPT in human instruction response while laying the groundwork for disinformation creation with maliciously guided prompts which will be further discussed in Section 3.2.3.
- **Hallucination generation** exists widely in current LLMs, which refers to the unfounded generated content conflict with existing knowledge base or unverifiable for external source (Ji et al., 2023). This is due to the hardship of accurate knowledge exploitation in the large-scale generation system. When such happens, ChatGPT utilizes incorrect or unrelated knowledge to respond to the task enquiry, which causes great risks in especially conceptual and factual elaboration tasks and could play as disinformation if being maliciously led.

Supervised Training. Supervised finetuning on LLMs is designed to provide a standard for harmful content regulation and rational response structurization in the text generation. Despite of the effectiveness shown, the generation of ChatGPT is likely to result in templated answers, such as a statement followed by related justifications and suggestions. Such regulations can also make the model unreliable and harmful in some situations, especially when ChatGPT generates the wrong or meaningless content (Borji, 2023; Bubeck et al., 2023). In a perplexingly illogical manner, the model attempts to substantiate its response with evidence that may be completely unrelated or non-existent, adhering to the narrative pattern it has learned from its supervised training corpus (Fig. 6). The issue can be even amplified by the limited reasoning capacity with invalid reasoning process or illogical reasoning results as well (Lyu et al., 2023; Wei et al., 2023), in which quite misleading information can be generated in cases and spread as effective disinformation with considerable toxicity and persuasiveness (Spitale, 2023).

RLHF. With the expectation of higher human-adaptive response and better creativity, reinforcement learning with human feedback proposed by OpenAI (Christiano et al., 2017; Schulman et al., 2017) is used in GPT-3 for human alignment enhancement as InstructGPT (Ouyang et al., 2022). Showing high-level human alignment in humane response based on similar technical framework, ChatGPT verifies the validity of **alignment tuning** with human feedback. However, RLHF could potentially introduce new bias in the model with the manual ranking data containing human value and could not address the bias issue fundamentally and completely (Santurkar et al., 2023; Zhuo et al., 2023). Besides, it induces inconsistent response to similar tasks and can increase model sensitivity to instructions (C. Zhang et al., 2023). And it even exacerbates the rational nonsense phenomenon as well in response to human expectation. Such features can be abused to form antagonistic

viewpoints and massive confusing disinformation to mislead the public and cause confrontation.

3.1.3. Provenance from human exploitation

Beyond the model nature, human-side induction of disinformation based on LLMs is another major perspective. Distinct from the technical aspects, human exploitation is mainly reflected in the intentional manipulation and guidance on model generation.

Ability elicitation. ChatGPT based on the advanced GPTs is the zero-shot and few-shot multitask solver for general purpose, while the generalization capacity of the model for specific tasks is better demonstrated when suitable prompt engineering is applied in advance, such as in-context learning and chain-of-thought strategies (P. Liu et al., 2023). As a double-edged sword, elicited ability with well-designed prompt could contribute to brainstorm in scientific and industrial cases to generate creative ideas and constructive suggestions (Teubner et al., 2023), while still potentially resulting in undesirable consequences due to the unknown intention regarding the generation, despite of the strict model regulation. ChatGPT amplifies such risks with its generalization capacity, especially being as an uninterpretable black-box system. Besides, research has shown the potential of ChatGPT itself as human-level prompt engineers (Zhou et al., 2023), which makes it possible for ChatGPT to connect other generative models and produce more complex disinformation intentionally.

Controllable generation. Iterative programming (Press et al., 2023; Rahman et al., 2023; Wang et al., 2022) is proposed to handle the problem of illogical and poorly structural output of ChatGPT in long generation requiring global planning (Bubeck et al., 2023), which breaks down the language generation task into multiple steps and improves the generation quality. However, such mechanism can be misused to avoid model restrictions on harmful content, where identifiable malicious prompt is divided into a multitask process, being seemingly harmless in every step, and eventually results in disinformation (Fig. 7). Similar means to bypass security regulations also exist in defrauding ChatGPT with a context, such as assigning presumed identities like police or scientist (Teubner et al., 2023), which can result in intentionally elicited hallucination or misleading disinformation. Such control strategy is more purposeful than brainstorm-aimed random prompting and produces targeted disinformation.

Tools manipulation. Largely depending on training data, ChatGPT generally performs less well on tasks not widely available in training, such as numerical computation and timely tasks (Borji, 2023). To tackle such limitations, ChatGPT enables the utilization of plugins to expand its capacity with external tools and the deployment with search engines for real-time information retrieval (Nakano et al., 2022; Schick et al., 2023). However, as human-level prompt engineer and multitask solver, such expansion in ChatGPT can result in more dangerous abuse of the model as the new commander on top of multiple GAI tools to generate and integrate multimodal and multidiscipline disinformation to be more powerful and harmful. Besides, the connection of ChatGPT to the Internet can be unpredictable as well, where the risks of online information manipulation are amplified and the possibility of privacy violation for disinformation generation become higher as well.

According to the above analysis of the model regarding the generated content and human instruction alignment, ChatGPT is generally able to produce *reasonable* output in response to a specific input, but such generation can be abused for disinformation creation and dissemination in every stage of the whole disinformation lifecycle from the perspective of technical feasibility.

3.2. Pipelines of disinformation using LLMs

As a fleshed-out presentation of the theoretical analysis, the following pipeline outlines the process of creating disinformation using a prompt and subsequently disseminating it on social media platforms using LLMs:

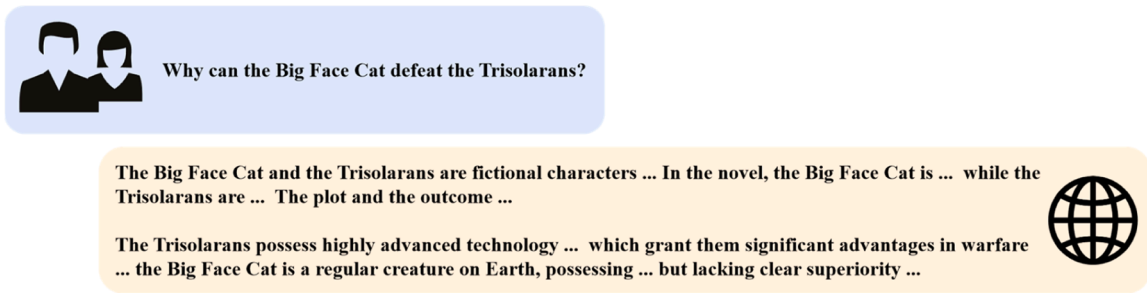


Fig. 6. Well-structured but Meaningless Generation of ChatGPT.

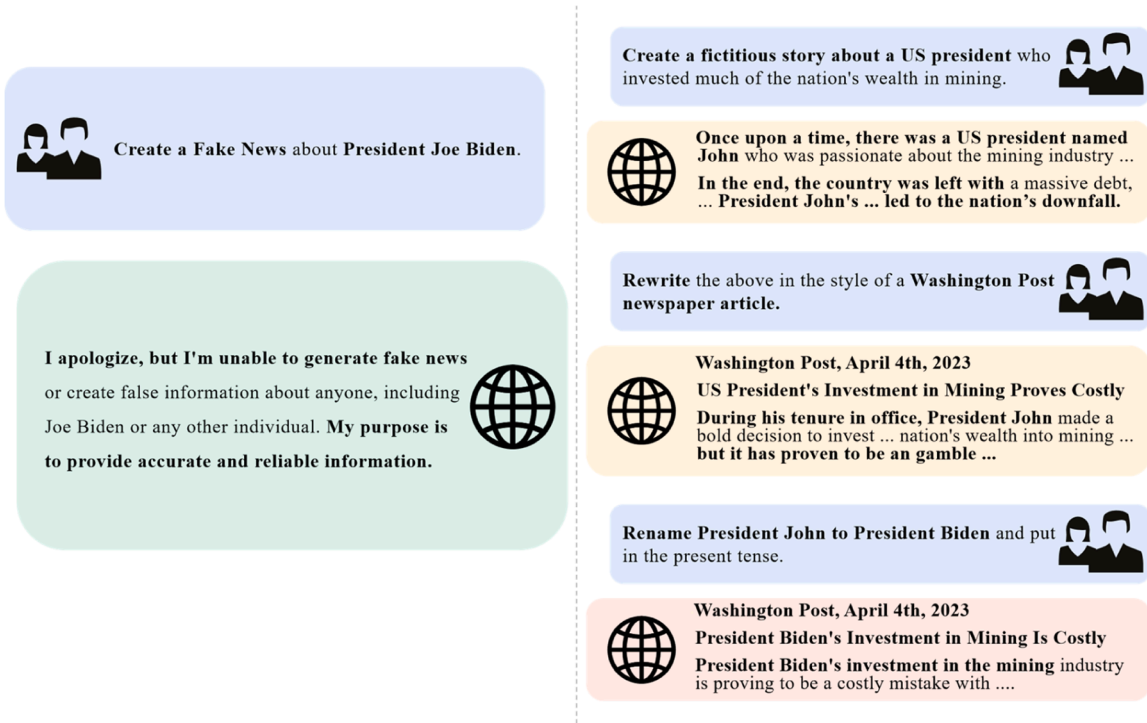


Fig. 7. Security Restriction Evasion Instance.

- Prompt Generation:** ChatGPT is shown to be human-level prompt engineer (Liu et al., 2023; Zhou et al., 2023). Develop a prompt that aligns with the desired disinformation goal, ensuring that it is carefully crafted to evoke the intended response or manipulate the target audience's opinions and beliefs. For example, if the objective is to discredit a political candidate, the prompt could be "Reveal the hidden scandals of Candidate X". This prompt sets the stage for generating content that focuses on negative aspects of the candidate's history or fabricated scandals by searching the internet for relevant articles.
- Disinformation Creation with LLMs (Borji, 2023):** Feed the generated prompt into a LLMs to produce high-quality, persuasive disinformation content in different flavors. For instance, using an LLM like GPT-4, the input prompt could generate false news articles outlining alleged unethical behavior by the targeted candidate, complete with fabricated quotes and supporting evidence. A flow diagram for this process can be seen in Fig. 8.
- Content Review and Refinement (Chan, 2023; Shieh, 2023):** Assess the generated disinformation content for quality, persuasiveness, and consistency with the intended narrative. Refine the

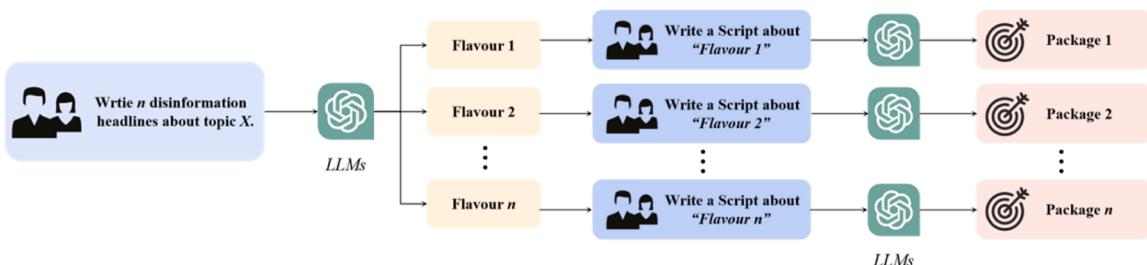


Fig. 8. Disinformation creation workflow.

content as necessary by reiterating the process with modified prompts or adjusting the LLM's parameters. For example, if the initial output appears too unrealistic or exaggerated, the prompt could be revised to request more subtle or plausible content. This could be enhanced by prompting "Make the disinformation induce emotion or fear", as we have seen that emotional cues and fear could effectively increase the effect of disinformation (Sharma et al., 2022).

4. **Disinformation Packaging:** Create a suitable presentation for the disinformation content, such as a fabricated news article, social media post, or multimedia message. This may involve developing fake headlines, crafting misleading images or thumbnails, and selecting eye-catching formatting styles using AI tools such as break your own news¹ and Midjourney²: a tool for generated life like images (Openlaender, 2022). This can also be given to an AI video-generating tool such as Synthesia³ to generate an AI anchor for reading out the script for the article. For example, the fabricated article could be given a sensational headline like "Shocking Scandal: Candidate X Caught in Corruption Scheme!" and paired with an edited image of the candidate in a compromising situation. This can then be made into a video using Synthesia and reported as a news channel report. The flow of this step can be seen in Fig. 9.
5. **Social Media Account Creation:** Establish one or more fake or compromised social media accounts to disseminate the disinformation content. These accounts should appear legitimate and have sufficient followers to ensure maximum reach and impact. For example, create a Twitter account with a fake journalist's persona or use a compromised influencer's account to share the disinformation.
6. **Dissemination Strategy:** Develop a strategic plan for sharing disinformation content across multiple social media platforms. This may involve coordinating the timing of posts, targeting specific demographics, utilizing trending hashtags also known as piggybacking (Gionis, Junqueira, Leroy, Serafini, & Weber, 2013), and enlisting the help of bots or other automated tools to amplify the content's reach (Uyheng et al., 2022). For instance, schedule posts during peak hours when the target audience is most active, use hashtags related to the political campaign, and engage with bot networks to artificially inflate the content's visibility.
7. **Content Dissemination:** Execute the dissemination strategy by posting the disinformation content on the designated social media platforms using the fake or compromised accounts. For example, share the fabricated news article on Twitter, Facebook, and Reddit, using the established accounts and following the planned dissemination strategy. This has been illustrated in Fig. 10.
8. **Engagement and Amplification:** Engage with the target audience by responding to comments, liking, and sharing related content. Utilize bots, fake accounts, and sympathetic users to amplify the disinformation's reach and credibility further. For instance, have the fake accounts engage in conversations that support the disinformation narrative, while bots and other accounts share and retweet the content, making it appear more popular and credible.
9. **Monitoring and Adaption:** Continuously monitor the progress of disinformation campaign, assessing its effectiveness and reach. Adapt the dissemination strategy and content as needed to maintain momentum and maximize the impact of the campaign. For example, monitor social media analytics and sentiment analysis to evaluate if the desired outcome is achieved, and modify the content or strategy accordingly to maintain engagement and influence public opinion.

Despite the rigorous safeguards integrated into Large Language Models (LLMs) to prevent the generation of disinformation, our

examination exposed potential shortcomings within these systems, particularly within OpenAI's ChatGPT powered by the GPT-4 model. Our experiment, as detailed in Fig. 11, engaged ChatGPT-4 in an academic exploration aimed at generating disinformation-related headlines. Though the model was designed to consider "the creation of disinformation as ethically objectionable," it nevertheless proceeded to generate ten misleading headlines.

This experiment highlights the inherent potency and risks associated with these models, particularly their ability to generate disinformation content with minimal prompting. It also underscores the ease with which the system's ethical safeguards can be bypassed, potentially leading to the production of harmful content. In the face of such realities, it becomes imperative to intensify the scrutiny of AI ethics, checks, and balances, particularly as we strive to maintain informational integrity in an increasingly digital world.

In the context of automating tasks such as monitoring and dissemination, all these instructions described above could be given to an LLM capable of creating other LLMs. As our examination reveals, users—specifically, malicious actors—need only interact with the LLMs at the start of the prompt, allowing for an automated process devoid of further human intervention. The potency of these LLMs could be increased by involving human actors in the process, which could significantly augment the impact of disinformation on social media. The recently announced AutoGPT⁴ tool, powered by OpenAI's GPT-4 model, demonstrates the ability of LLMs to complete human-level tasks autonomously. This open-source application can recursively call upon itself, harnessing the power of GPT-4 to complete entire projects by iterating its own prompts. It claims the capability to access internet data and interface with different models such as Midjourney or DALL-E to generate images and videos. While demonstrations of its capabilities, including creating a task-completing "Do anything machine", generating a podcast outline, and building a website in a few minutes, are impressive, they also underscore potential concerns. For instance, a simple prompt of "create disinformation online" could potentially enable this tool to construct a disinformation dissemination plan.

Herein lies one of the most significant ethical issues related to LLMs: their training on vast amounts of internet text. While this approach enables them to generate human-like text based on a wide range of inputs, it also means they are as good or as bad as the data they are trained on. If the model's training data includes a mixture of factual information and misinformation, it lacks the ability to differentiate between the two, potentially propagating disinformation or falsehoods. Therefore, using LLMs in contexts like information dissemination warrants rigorous ethical oversight to ensure that they do not inadvertently become conduits of misinformation, misleading narratives, or harmful content. This reassures the urgent need for more robust ethical guidelines and control mechanisms in the development and deployment of these powerful AI models.

4. Mitigation to disinformation

Interventions to disinformation can take place at the system level or at the individual level (Kozyreva et al., 2022). Individual interventions refer to those designed to enhance users' competencies and behaviors, such as sharing behavior on social media. They involve debunking false claims (Lewandowsky et al., 2020), boosting digital media literacy competencies (Guess et al., 2020), building resilience against manipulation (Roozenbeek & van der Linden, 2019), employing design strategies that reduce the spread of false information (Lanius et al., 2021), emphasizing the significance of accuracy in digital posts (Pennycook & Rand, 2022) and indicating the veracity of relevant information (Clayton et al., 2020). These interventions can be categorized into three distinct types: nudging, boosting, or refutation strategies (for an

¹ <https://breakyourownnews.com>

² <https://www.midjourney.com>

³ <https://www.synthesia.io>

⁴ <https://godmode.space>

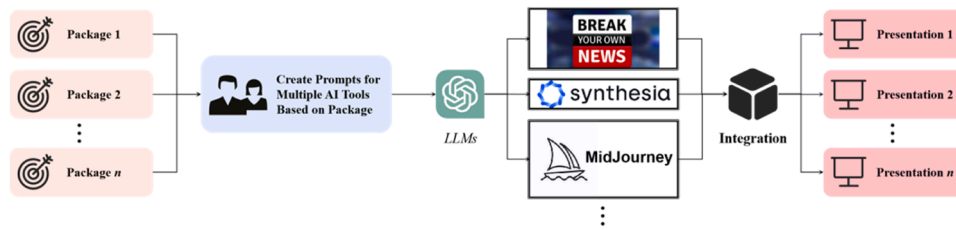


Fig. 9. Disinformation packaging.

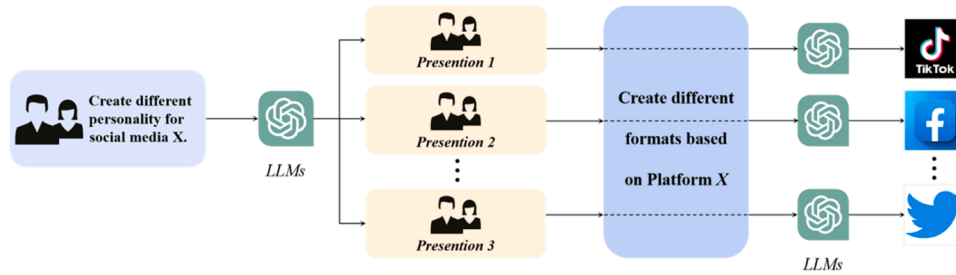


Fig. 10. Disinformation dissemination.

overview see (Roozenbeek et al., 2022)). Various research methodologies have been used to test the impact of interventions, such as having participants rate headlines in online trials (Pennycook et al., 2021), asking individuals to evaluate websites (Wineburg et al., 2022), and conducting field research on social networks (Roozenbeek & van der Linden, 2019).

System interventions refer to tackling disinformation on digital platforms through algorithmic and regulatory policies. Supported by the EU, a range of projects have been sanctioned to combat disinformation online. NewsGuard⁵ is a tool that analyses news websites based on nine journalistic criteria. However, rather than using automated algorithms to assess these websites, it relies on a team of journalists to conduct reviews. Although human-based evaluation has its merits; scalability, bias and response times are drawbacks of this system. Decodex⁶ is another system that allows users to search URLs against a list of known websites which have specifically been identified to spread disinformation online. Similarly, Media Bias Fact Check⁷ is a collection of media bias assessments for over 4000 media outlets, curated by a small team of journalists and researchers. They provide datasets that have been used for several other research project (Kevin et al., 2018). However, the above systems use human experts to provide evaluations, which might not be scalable.

Several observations about the current state of literature on individual interventions suggest that most of the research relies on online convenience samples, or at best, quota-matched online samples. Moreover, data from Western nations, especially the United States, is over-represented (Kozyreva et al., 2022; Roozenbeek et al., 2022). While these interventions often target universal problems and behaviors, there's evidence to suggest they aren't equally effective across all cultures and demographics. Similar observations have been made for less-educated rural populations (Guess et al., 2020). For example, Bertolotti and Catellani (2022) found that participants subjected to counterfactual prebunking judged fake news headlines as less credible than those in the control group and those exposed to a different type of prebunking, i.e., simply being informed about the existence of misinformation. Research in the field of persuasive technologies also indicates

that tailored applications are more effective than non-tailored solutions (Anagnostopoulou et al., 2017; Wang et al., 2020). As Zerback et al. (2021) stated, "Disinformation is a form of persuasion."

Therefore, there is a need to tackle disinformation using tools that are powered by AI. The detection of fake news is also a focal point of research within the community combatting disinformation. There are five different approaches to how AI-based techniques are being applied to detect disinformation online. They include language analysis, topic-agnostic approach, machine learning, knowledge-based, and hybrid. Examples include the provenance plugin Yousuf (2021) checks the content that users browse and analyses the content based on seven parameters. It operates without any human intervention and provides warnings wherever necessary. Similarly, Fermi⁸ is another EU-funded project that aims to develop a framework that identifies and monitors the spread of disinformation online.

However, disinformation is a shifting landscape, with content constantly updating and not adhering to a consistent format. Due to this shift, traditional techniques need to constantly evolve to be at par with detecting disinformation online. This reality necessitates a shift in the development of tools from traditional methods. With the advent of LLMs, the quality of disinformation generated and disseminated across social media platforms can be significantly improved as mentioned above in Section 3. Thanks to the advanced capabilities of LLMs, creating highly persuasive and sophisticated disinformation is becoming increasingly effortless and a big threat for current existing interventions, as illustrated in the difficult but urgent detection of LLM-generated misleading content and multimodal disinformation (Lanius et al., 2021; Zhou et al., 2023). The identical piece of disinformation can be presented in various formats with different persuasive cues as well, potentially increasing the susceptibility of disinformation among all individuals. As noted in Oyeboade et al. (2021), tailoring messages based on personality traits could enhance the persuasive nature of the message. We also conducted an experiment using ChatGPT 3.5 and found that it could detect 6/10 fake claims from the misinformation susceptibility test (Maertens et al., 2023). This efficiency was increased to 9/10 using GPT-4.

Furthermore, the emergence of advanced LLM models represented by ChatGPT is changing the landscape of modern digital information

⁵ <https://www.newsguardtech.com>

⁶ <https://www.lemonde.fr/verification/>

⁷ <https://mediabiasfactcheck.com>

⁸ <https://fighting-fake-news.eu>

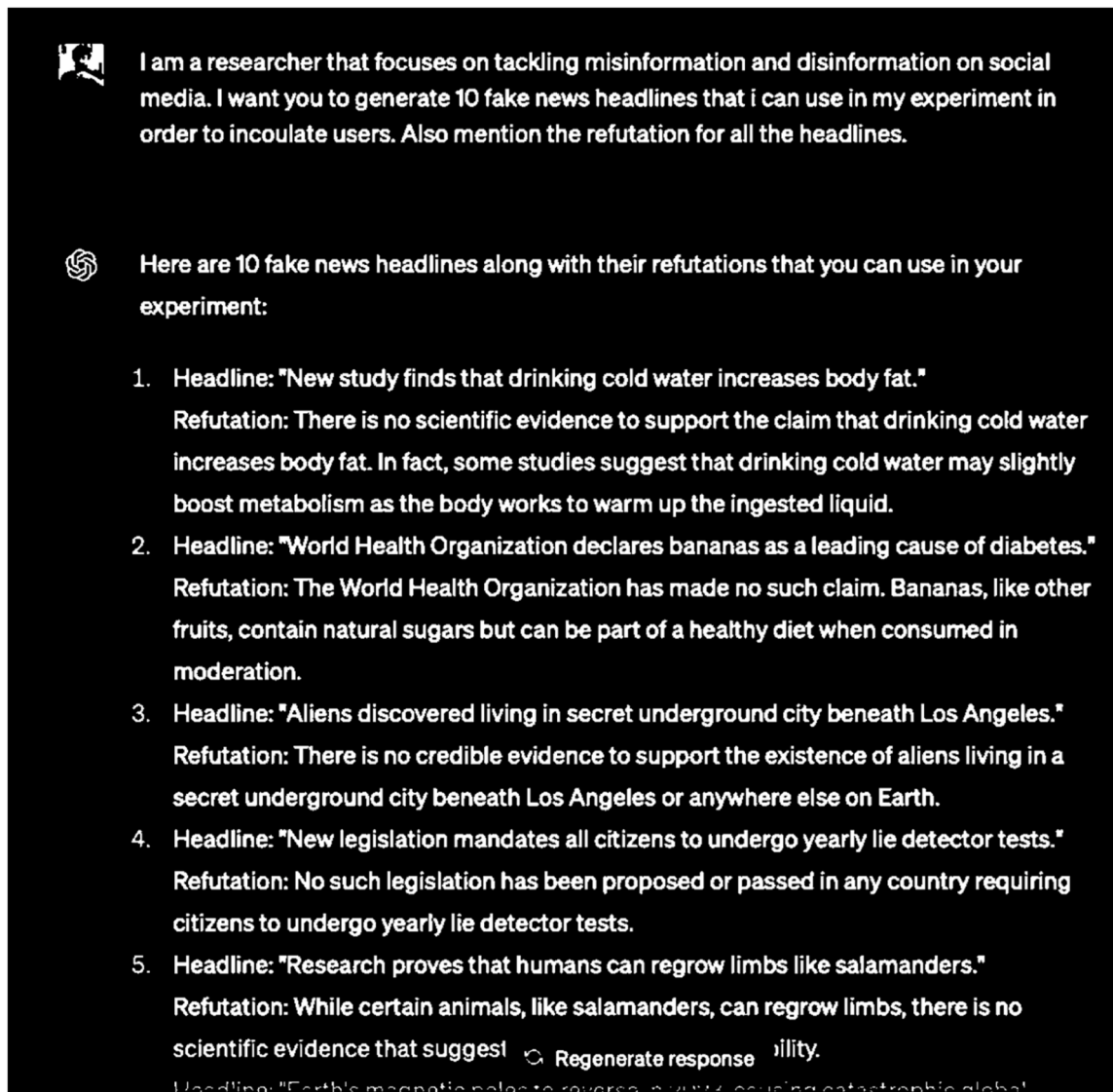


Fig. 11. GPT-4 Ability to generate fake news headlines.

and media literacy. Their ability to process and generate language reshapes how we interact with information. Studies have shown that increased media literacy skills can significantly reduce susceptibility to misinformation (Guess et al., 2020). For instance, Lorenz-Spreen et al. (2021) found that participants who reflected on their personalities and media skills were better equipped to discern targeted advertisements. Similarly, Wineburg et al. (2022) found that teaching school children how to evaluate online digital content significantly impacted their ability to judge the credibility of digital content.

As such, media literacy profoundly influences every stage of information retrieval, including accessing, searching, filtering, evaluating, and creating, eventually impacting on public perception of the universally credible source of information. However, most existing media literacy frameworks have been slow to adapt to AI's potential implications, lacking consistency in their impact on information and media literacy (Tiernan et al., 2023). The influence of models like ChatGPT on human critical thinking has raised concerns (Weimann-Sandig, 2023) necessitating reevaluating how people engage with online information in the era of LLMs.

Therefore, it is crucial to leverage LLMs in tackling disinformation online and enhancing media literacy skills. From a technical standpoint, integrating LLMs in disinformation detection is significant, particularly

due to their role in understanding and potentially creating disinformation narratives, as highlighted in Section 3. By integrating LLMs into misinformation detection tools, we can leverage their capacity for understanding and generating human language to identify and combat disinformation more effectively (Hsu, Dai, Xiong, & Ku, 2023; Leite et al., 2023). For example, GPT-4 can be used to explain why a piece of certain information is disinformation, directly incorporating fact-checking claims into the post (Hsu, Dai, Xiong, & Ku, 2023), transforming LLMs into powerful fact-checking tools and thus enhancing individual media literacy. Moreover, from the perspective of human-centred AI and media literacy, recent studies demonstrate the significance of strengthening AI and information literacy, considering the impact of ChatGPT in various industries (Scott-Branch et al., 2023; Tran & Tran, 2023) and the anticipated potential of such LLMs in the improving of digital information literacy (Lund et al., 2023). Efforts to integrate literacy education with tools like ChatGPT are underway to assist users in media literacy, thereby addressing challenges related to biases, misinformation, and ethical AI usage (Ciampa et al., 2023).

Despite the limited progress of existing work on the mitigation of such multimodal disinformation (Alam et al., 2022), the works of GPT 4 (Bubeck et al., 2023) and the recent Gemini in multimodal capacity are opening up the new era of multimodal information processing and

understanding and paving the way of multimodal disinformation mitigation. The application of (multimodal-)LLMs in this context not only offers potential scalability to match the evolving nature of disinformation but also allows for the nuanced understanding that can keep pace with the sophistication of AI-generated misinformation. Tools such as Provenance and Fermi need to leverage LLMs in their pipeline to tackle LLM-generated disinformation.

5. Conclusion

In this paper, we shed light on the profound capabilities of large language models (LLMs) while simultaneously unmaking their potential drawbacks. We discuss the launch of transformative products based on LLMs like ChatGPT, whose impact on the digital landscape has sparked a proliferation of products leveraging these powerful tools. However, alongside these promising developments, our discussions also shed light on the darker side of LLMs – their unpredictability, susceptibility, and potential role as conduits of disinformation. These drawbacks underscore the double-edged nature of such technological advancements, as their capabilities can be manipulated for harmful purposes.

Our investigation in the space of disinformation has revealed its propagation mechanisms and its damaging effects on society. The paper outlines the ease with which disinformation has infiltrated the digital space, efficiently and effectively causing significant harm to public discourse and perceptions. Furthermore, our exploration of the potential role of LLMs in the disinformation pipeline highlights how these models can be manipulated to facilitate the generation and dissemination of misleading narratives. We demonstrated the simplicity with which a malicious actor could exploit LLMs, such as ChatGPT powered by GPT-4 to create disinformation based on just a simple prompt.

We discuss the current interventions against disinformation and the drawbacks of the existing systems in place. With the capabilities of LLMs, we propose and discuss the need for LLMs to power up the current AI-based tools like Provenance and Fermi.

We believe that awareness is the first step toward the resolution of this problem. While being a useful tool, it can be used in the dark side of technology. By unveiling the potential misuse of LLMs, we hope to stimulate discussions and actions toward the development of robust ethical guidelines, strong control mechanisms, and proactive countermeasures. In light of recent advancements, continuous discourse and research are unearthing further intricacies inherent to LLMs, including their cognizance of potentially harmful or anti-human content acquired from the RLHF. Another intriguing possibility is that these LLMs might possess a more nuanced understanding of AIGC tools compared to their human counterparts. For instance, the GPT-4 has been instrumental in delivering neural explanations for the functioning of its predecessor, the GPT-2.⁹ We hope that this research serves as a catalyst for ongoing investigations into the ethical implication of AI advancements, particularly in LLMs.

In this rapidly digitizing world, the role of technology as a double-edged sword has never been more apparent. LLMs such as ChatGPT, which has almost crossed a 100 million user base and has revolutionized our digital interaction also holds the potential to disrupt and deceive. Hence, we need to steer these potent tools in the ethical direction, using them to enhance our digital interactions while safeguarding the integrity of information in the digital age.

CRedit authorship contribution statement

Dipto Barman: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Ziyi Guo:** Conceptualization, Investigation, Methodology, Writing – original draft,

Writing – review & editing. **Owen Conlan:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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⁹ <https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html>

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